

Project Brief

CanPlan 2.0: Cloud-Based Expansion

What is CanPlan?

CanPlan is a versatile visual task management and executive functioning mobile app. It helps users break down tasks into easy-to-follow steps using photos, captions, speech, and video recordings, making it ideal for anyone who needs support with task sequencing, to-do lists, and planning and staying on track to complete a task successfully.

CanPlan allows you to use your real-world tasks in an environment where they are completed. The task builder is flexible and intuitive, with no icons or prescribed sequences. Just a blank canvas to create your own tasks!

Who is it for?

Anyone who can benefit from accessing step-by-step instructions to complete a task successfully and at the right time.

CanPlan is designed specifically for individuals with acquired brain injuries, early-onset dementias, youth with cognitive or behaviour barriers and workers with diverse abilities.

Current state of development and availability

In its current state, CanPlan operates exclusively on local storage, ensuring that all user data remains on the original device. While this architecture provides a simple, robust, and secure environment that is fully functional without an internet connection, it creates significant barriers to scalability and feature expansion. Because tasks are tied to a single physical device, users lack the ability to sync their progress across multiple platforms or share data with their support networks.

Although the current version is built using the Flutter framework and is available as a free download for both iOS and Android, the reliance on a localized database limits the app's potential.

Our Vision for CanPlan 2.0

The next generation of CanPlan will build on the success of our original app. We're keeping the simple, easy-to-use interface while adding support for user accounts and cloud storage. This new version will open the door to a richer, more integrated experience for users and the people who support them.

Key features for CanPlan 2.0

- **Individual User Accounts:** Log in from any supported device to securely access your content, which is stored in the cloud.
- **Offline Mode:** All your content is available even when you're not connected to the internet.
- **Support Person Connectivity:** A teacher, employer, or other support person can connect their account to yours. This allows them to create and share tasks, check in on your progress, and communicate with you directly through the app if you need help with a task.
- **Remote Task Management:** A centralized portal that allows supporters to remotely build, edit, and "push" customized task sets directly to individual users or entire groups.
 - **Progress Tracking & Telemetry:** The dashboard is designed to record task initiation and completion data, providing a high-level view of how users are interacting with their tasks.
 - **Reporting:** It includes functionality to convert that telemetry data into professional PDF reports for clinical or educational review.
- **Real-Time Connectivity & Support:** A comprehensive "just-in-time" assistance suite featuring, in-app messaging, and text support that connects the user's mobile app directly to a supporter's device to help navigate barriers and provide discreet, real-time guidance during task execution.
- **Bluetooth Switch Accessibility:** The app will be compatible with alternative input devices, such as Bluetooth accessibility switches, to support individuals who can't use a mobile device's touch screen.
- **Terminal Mode:** This mode locks down the mobile device, allowing the user to access only the CanPlan app. This is great for situations where the full functionality of the device might be a distraction or a barrier for the user.
- **Use case modules and setup wizard:** CanPlan's next generation will feature customizable modules designed for specific user groups. A setup wizard will help new users optimize their experience to align with their unique needs.
 - **Employment:** This module focuses on job success, helping users with time management, preparing for the workday, managing shift schedules, and receiving timely notifications.
 - **Education:** Designed for the classroom, this module supports educational curricula and helps younger users build independence and autonomy.
 - **Seniors and Dementia:** CanPlan can be customized for each user's needs. It will give simple reminders, like "It's time to drink a glass of water," and help people follow the steps of more complex daily tasks. A new feature will also let a family member or

caregiver record their own voice to send personal reminders whenever they're needed.

New Capabilities and Feature Ideation

During the development process, the CanAssist's team will serve as "Voice of the Client" to ensure that the platform remains truly user-centric, the team will lead community-led research and engagement across a broad spectrum of ages and abilities. This includes active consultation with both prospective users and their support networks to gain insights into unmet needs, gather feedback on proposed solutions, and ensure that iterative development is grounded in lived experience.

While this engagement directly guides core feature determination, there is a strong emphasis on open innovation. We welcome new feature ideation and capabilities that push beyond the application's current design and technical constraints.

Technical Requirements

- Android and iOS Versions – platform-agnostic design for all device compatibility
- Scalable - Ensure stability and responsiveness to 10k+ users
- Meet a high level of accessibility, achieving WCAG (Web Content Accessibility Guidelines) standards
- Offline functionality
- Portrait and landscape support
- User account creation and password reset support (potentially SSO capability)
- Privacy and security:
 - FIPPA compliant
 - AES256 Encryption standard
 - Documented penetration testing (outside of class project scope)
 - UVic PIA and systems privacy and security consultation (outside of class project scope)

Project Deliverables

- Conduct a comprehensive a feature and capability evaluation of CanPlan 1.0
- Define and document user stories that capture the diverse needs of both primary users and their support networks to guide the product's design and engineering effort.
- SRS document that specifies the functional and non-functional requirements, including cloud architecture, security protocols, and accessibility standards.
- Development platform selection – make an informed decision and recommendations for a frontend development language and environment.
- Backend platform selection – make an informed decision on the backend codebase language and platform.
- Determine hosting requirements and specify a server architecture
- Mobile App Implementation
- Codebase Documentation
- Handoff for continuation support

App Development Roadmap

To support the practical development of the application within the context of a student-led capstone team, the development process should follow a modular path. Each stage will focus on achieving critical features and functionality that can be tested in real-world circumstances and environments.

This approach ensures that the project scope and deliverables at each milestone result in a usable, testable product without overcommitting the student team. The proposed modules support the initial framework and architectural creation, front-end development (including accessible UI/UX), and back-end coding and hosting implementation.

Depending on project team size modules can be allocated to multiple teams simultaneously or across multiple semester or course session.

Module 1: Technical Framework Specification and UI/UX Design

- **Technical Architecture:** Define the full technology stack and backend infrastructure.
- **Requirement Analysis:** Perform complete feature discovery and a formal Software Requirements Analysis (SRA).
- **UI/UX Design:** Develop fully scoped page views and interface elements.
- **Accessibility:** Implement advanced accessibility best practices and compliance standards.
- **Front-End Development:** Complete build-out of all user-facing features and interface components.

Final Deliverable: Proof of concept release and documentation package.

Module 2: Core Development

- **Backend Development:** Complete build-out of server-side logic and core functionality. **Architecture & Hosting:** Design of the server environment and selection of hosting infrastructure.
- **Security & Privacy:** Implementation of data protection protocols and privacy compliance.

- Scalable Storage: Setup of flexible data and file storage solutions.
- Integrations: Full integration of third-party products

Final Deliverable: Functional and tested back end.

Module 3: Integration and Production Release

- System Integration: Full integration of the front-end with the back-end API and hosting infrastructure.
- Quality Assurance: Comprehensive software testing to ensure reliability and accessibility.
- Performance: Regular measurement of system speed, responsiveness, and load handling.
- Operations: Planning for ongoing maintenance, security updates, and long-term support.

Final Deliverable: Production release.